

Soln: The three planes $x_1 + 2x_2 + x_3 = 4$, $x_2 - x_3 = 1$, and $x_1 + 3x_2 = 0$ and ~~are~~ have at least one common point of intersection if and only if the equations are consistent.

Augmented matrix: $\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & 0 \end{bmatrix}$ Row 3 = Row 1 - Row 3

Row 3 = Row 3 + Row 2

$$\begin{bmatrix} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

The last equation is $0=5$. This is a contradiction
 $\therefore$ The equations are inconsistent and there is no common point of intersection.

In Exercises 19-22 determine the values of h such that the matrix is the augmented matrix of a consistent linear system.

19) $\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix}$ Row 2 = Row 2 - 3xRow 1

$$\begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix}$$

$$x_1 + hx_2 = 4$$

$$(6-3h)x_2 = -4$$

$$\Rightarrow x_2 = \frac{4}{3h-6}$$

$$x_1 + h\left(\frac{4}{3h-6}\right) = 4$$

$$\Rightarrow x_1 = 4 - \frac{4h}{3h-6} = \frac{8h-24}{3h-6} = \frac{8(h-3)}{3(h-2)}$$

For the system to be inconsistent, $6-3h=0 \Rightarrow h=2$

$\therefore h \neq 2$ means the system is consistent $\square$

2) $\begin{bmatrix} 1 & 3 & -2 \\ -4 & h & 8 \end{bmatrix}$ Row 2 = Row 2 + 4xRow 1

$$\begin{bmatrix} 1 & 3 & -2 \\ 0 & h+12 & 0 \end{bmatrix}$$

$$x_1 + 3x_2 = -2$$

$$(h+12)x_2 = 0$$

$$\Rightarrow x_2 = 0 \text{ or } h = -12$$

The system is consistent for all h $\square$